

Project Report

Wildlife Information System (WIS)

Document Number: WIS-SRS-001
Version: 2.0
Date: June 2026

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1. INTRODUCTION

1.1 Background

Wildlife means animals that live naturally in forests, mountains, deserts and in other environments like elephants, lions, wolves, dogs etc. These animals are very important because they help to maintain the balance of nature. For example: predators such as wolves control the population of other animals, and the birds help plants to grow by spreading seeds and pollinating flowers. Now, much information about wildlife is collected around the world. This information includes Animal, population number, Movement and migration routes, Animal behavior, DNA and genetic information, Health and disease records. However, the data is stored in many different places. Different organizations, universities, researchers, and government departments collect data separately. There is no single system where all wildlife information can be found. In Pakistan and many other countries, wildlife data is usually stored in reports, spreadsheets, notebooks, and paper records. Because of this, it becomes difficult for researchers, Doctors, and government agencies to access and use the information.

To solve this problem, Our System the Wildlife Information System (WIS) is proposed. WIS is a centralized system that stores all wildlife data in one organized database. Users can easily access and share information through an API, making wildlife research and conservation more efficient.

1.2 Existing System

Currently, wildlife data in most countries is managed in one of the following ways:

- Government forest departments maintain their own paper base or basic spreadsheet or excel file records of wildlife populations and incidents.
- Research universities collect data during research or field trips and store it in local databases that are not accessible to the public.
- International conservation organizations like WWF and IUCN maintain global databases, but they are limited and they have not included local or regional level details.
- Some IoT based systems track animal movement using GPS collars, but this data is kept private and not shared in public platforms.

The major weakness of all these approaches is that they do not talk to each other. There is no standard system that collects, organizes, and shares wildlife data openly.

1.3 Issues & Motivation

The following problems motivate the development of the Wildlife Information System:

- **Data Isolation:** Wildlife data exists in silos. Researchers, forest departments, and NGOs each have their own records but cannot share them.
- **No Standard Format:** Data is stored in different formats like Excel files, PDF reports, handwritten notes etc. we are making it impossible to combine or compare across organizations.
- **Lack of Open Access or Source:** Most wildlife databases are closed. Researchers must physically contact organizations and wait weeks to get data they need for their studies.
- **Missed Opportunities for Analysis:** Because data is fragmented, important patterns such as disease outbreaks, migration route changes, or population decline are often noticed too late.
- **Waste of Research Resources:** Multiple researchers sometimes collect the same data independently because they do not know it already exists elsewhere. Which waste their time.

1.4 Problem Statement

There is no single system in Pakistan where all the wildlife information is stored or shared. The different universities, research institutions and government institutions have but they keep their own records separately or private, because of this. It is most difficult to find information about animals like their populations, their behavior, DNA and Their Health. This creates many problems Researchers may collect same data repeatedly because they don't know it already exists. And also, the most important information can be missed, and govt agencies may take decisions without complete or accurate data.

The Wildlife Information system solves these problems. It will store wildlife information in one central database. And Authorized users can access this information through secure API and they can use it for research, wildlife monitoring, conservation projects and they can use for decision making.

1.5 Scope

The Wildlife Information System will cover the following:

- It will keep all the wildlife information in one central database. The information will include like animal profile, Population, movement tracking, DNA records, health, their behavior and environment data.
- There will be a backend system that will validate, organize, and manages all incoming data from different sources, like field researchers, IoT tracking devices, and lab results.
- It will include an API. A RESTful API layer that allows external users and applications to send queries and receive structured wildlife data in standard format like JSON.
- The User roles include Data Entry Operators, Researchers, Government Officials, and System Administrators, each will have different level of access.
- There will be a basic web interface in which can search and can enter data manual.

The following are explicitly outside the scope of WIS:

- This system will not install GPS collar or tracking device, manufacturing, or design. It will only store data transmitted from existing devices.
- Real-time live animal tracking maps or animated movement visualization on a geographic interface (this may be a future Phase 2 extension).
- This system will not provide any real time data or tracking or animated maps showing animal movements. But this may be a future plan in version 2.0 extension
- This system will not provide any mobile application for android or iOS. But the other developers may build these separately using the WIS public API.
- This system will not use AI or ML for identifying animal from images or sensor data. This system stores records but does not perform intelligent pattern recognition.
- This system will not connect directly with international database like IUCN Red list or Cities in v1.0 but may connect in later versions.
- This system will not include financial, HR, payroll, or administrative management features. It's mainly focuses on ecological and animal data.
- This system will not include offline data collection tools or special devices for field use in remote areas where the internet is not available.
- This system will not provide automatic alerts for diseases outbreaks, environmental risks or illegal hunting etc.
- This system will not support multiple languages. this system will operate in English only in v1.0.
- The system will not handle legal reporting wildlife permits or enforcement tracking systems.

1.6 Project Management

This project will be managed using a structured approach with clearly defined phases, responsibilities, and timelines.

Phase	Activity	Duration
Phase 1	Requirements gathering and analysis	2 weeks
Phase 2	System design (DB schema, API design)	2 weeks
Phase 3	Backend development and database setup	3 weeks
Phase 4	API development and testing	2 weeks
Phase 5	Documentation and final review	1 week

The Tools used: GitHub for version control, WhatsApp and Discord for team communication, and Google Docs for shared documentation. Weekly meetings every Saturday to review progress.

1.7 Technological Overview

Component	Technology
Backend Framework	Node.js with Express.js
Database	PostgreSQL (relational, for structured data)
API Type	RESTful API (JSON responses)
Authentication	JWT (JSON Web Tokens)
Frontend (basic)	HTML, CSS, JavaScript
Version Control	Git / GitHub
Hosting (future)	Cloud (AWS or DigitalOcean)
Data Format	JSON and CSV for exports

1.8 Goals & Responsibilities

Primary Goal: To build a centralized wildlife information database that is accurate, well organized, and accessible through an API for research and analysis purposes.

Secondary Goals:

- Encourage data sharing among wildlife organizations.
- Provide a platform that supports conservation and research decision making.
- Build a scalable foundation that can grow with more data and users over time.

1.9 Process Model

This project will follow an Incremental Development Model. which means the system is built step by step approach, where each step adds new functionality and improve previous. This model was chosen because:

- Requirements are mostly clear, but small changes are expected during development.
- It allows early testing of core features like the database and API before the full system is complete.
- Each increment produces a working version that can be reviewed and improved.

The three increments are:

- Increment 1: Database design and basic data entry system.
- Increment 2: API development and user authentication.
- Increment 3: Search, filtering, export, and final testing.

1.10 Project Management System

The team will use GitHub Projects to manage the work of the Project. All Tasks will be organized in a board format with three columns:

1. **To Do:** The tasks that need to be started
2. **In Progress:** The tasks that are currently being worked on.
3. **Done:** The tasks that have been completed

Each team member is assigned specific issues or tasks on the GitHub repository. Which will help us in tracking progress and make sure every task is completed on Time.